



WPI

BIKI OPTIMIZATION

REPORT

2025

Presented By

WPI Bikeshare IQP Team



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1 – Preface

The purpose of this report is to outline key issues in Biki’s network and to recommend improvements based on our analysis. We recognize that many of these issues stem from budgetary constraints, but we present our findings in their entirety with the goal of supporting long-term planning and guiding future investments.

Sincerely,

Alex Fenwick, Mattan Mohel, Ryan Nemti, Philip Ostojic



2 – Community Rapport

A major theme from the free-response section of our survey was Honolulu’s enthusiastic cycling community. Many respondents expressed strong interest in using Biki’s services, yet they also reported feeling discouraged by the state of bike share infrastructure and maintenance/rebalancing. We’ve included standout responses below, both to highlight the challenges Biki is currently facing, and to showcase the strong community you’re building despite them:

“Bikes could be maintained better, taken care of by the community.”

— Downtown Biki User

“Biki is an amazing tool... yet the service has been declining significantly since COVID. Rates increased with the promise of additional bikes being purchased... Considering working with Hawai’i Bicycling League to get further support from the cycling community and better promotion. Encourage the City to invest!”

— Downtown Biki User

“Honolulu’s bike sharing has potential, but car-first infrastructure, fragmented routes, and limited protection make it difficult to use safely and consistently for everyday transportation.”

— Kaka’ako / Ala Moana Biki User

“Love Biki and hope that the city taking ownership means the system can grow without being beholden to a profit motive... Grateful to Secure Bikeshare for getting Biki off the ground and to this point and I hope the city follows through on all the vision and potential there is for Biki in Hawaii!!!

Biki is a thousand percent the reason I was able to go car-less in 2017 and I get so much JOY out of utilizing our multi modal transportation within our public transportation system!

....It’s so freakin’ good!”

— Diamond Head Biki User

It’s clear that Biki has many avid supporters who not only rely on, but *enjoy* using your services. While we present data-driven recommendations later in this report, we would be amiss not to emphasize our belief that leveraging this community is the key to your long-term success in Honolulu.

A bike share program is fundamentally a people-driven network, and word-of-mouth within the community is your strongest asset. We fully acknowledge the budget constraints Biki faces, and we wholeheartedly recommend community-building as one of the most promising avenues for sustaining and growing your services in the long run.

2.1 – Rebalancing

One of the biggest issues reported in the survey was a general lack of bikes at stations. Bike share services like Citi in New York City offer highly successful rebalancing programs called *Bike Angels* where users can take bikes from areas of low demand to areas of high demand to earn credit. Alongside programs for community-driven rebalancing, We recommend further research into software to predict bike demand in order to make Biki’s rebalancing strategic rather than reactive.^a

2.2 – Memberships

“I understand that the price is higher, but I think the plan needs to be improved. A 30-minute ride or a one-day plan is not worth it... a weekly plan is the best way to get tourists...”

— Visiting Biki User

Many respondents expressed concern over the new price of membership plans. A lot of concern was expressed on time limits, especially when considering issues with unregistered bike returns. Plans like *The Voyager* and *The Commuter* enforce price differences between 30-minute and 45-minute bike rides that feel arbitrary from the perspective of the user. We recommend further surveying into the consolidation of membership plans to reduce the amount of options and make price options more transparent.

2.3 – Everybody Rides

According to your latest statistics, there are less than 10 unique *Everybody Rides* users in total. Additionally, the process for applying for the program is hidden away at the bottom the Biki website. Encouraging people to apply to this program will support Biki’s brand as a community member and diversify your user base.

2.4 – Communication and Outreach

“My most used station was removed suddenly, although it was a popular location...”

— Downtown Biki User

After reviewing Biki’s social media, we’ve found that your pages garner minimal traffic with close to no community interaction. Giving Biki users more say on the development of the Biki network is key to building brand loyalty and to cultivating a good relationship with your riders. Some examples for community input are

- Survey Biki users who frequent a station prior to its relocation to gauge how the station’s removal would impact their access to your services
- Have Biki users vote on new station and bike lane locations
- Posting on social media more regularly and participating in more biking events

^aThere are new methods being developed for predictive modeling of bike demand using machine learning methods based on weather, time, and seasonality.

3 – New Station Locations

3.1 – Location 1: Waialae Ave

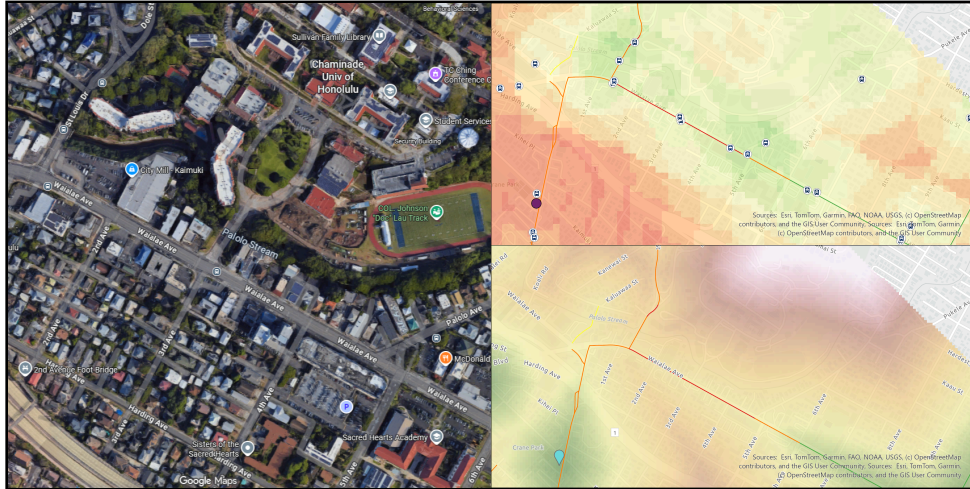


Figure 1: Google Maps(left), Suitability Map 1(top right), Bike Lane Connectivity(bottom right)

Our most suitable location is the region around the Chaminade University of Honolulu on Waialae Ave. This area is near a university so it can take advantage of the students that need to get further off campus. The university also creates a ‘mini downtown’ that has lots of different shops and points of interest. Furthermore, Waialae Ave has a bike lane and connects to two stations, 161 and 162, that are separated from the rest of the system.

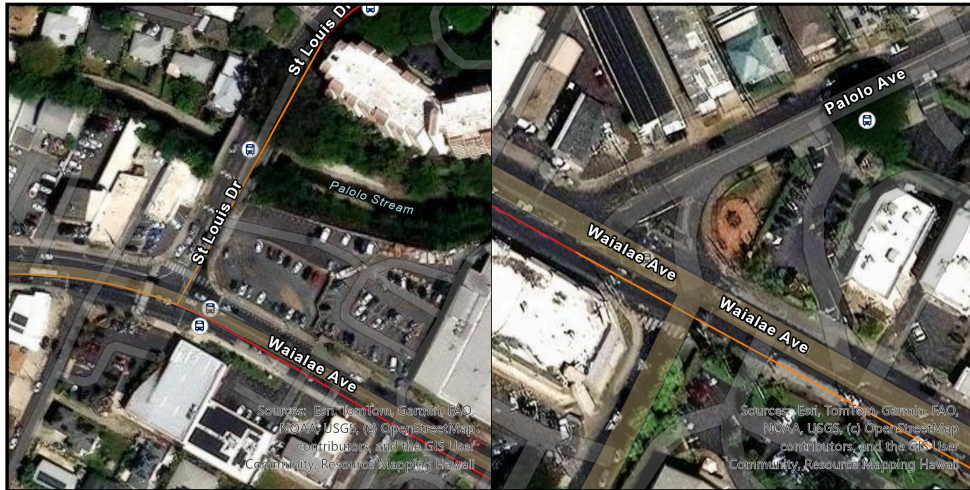


Figure 2: Waialae Ave / St Louis Dr(left) Waialae Ave / Pālolo Ave(right)

For this location we have two separate candidates for a new station, as seen in Figure 2. The first one would be at the intersection of Waialae Ave and St Louis Dr. This location is still on Waialae Dr while connecting to the neighborhoods that extend to the side of the university. The second candidate is at Waialae Ave and Pālolo Ave, at the Pālolo Triangle Mini Park. This spot is right in the middle of the ‘mini downtown’, being

right next to some restaurants and across the street from a shopping center. It could hopefully be integrated with the park, as it is operated by the city of Honolulu.

“I really wish there was a Biki station close to my home—say, at the corner of St Louis Dr. and Waialae Ave. Chaminade, St. Louis school, and City Mill are all in the same area and I would use the service much more often if I didn’t have to walk 10 minutes to get to a station.”

— Diamondhead Biki User

3.2 – Location 2: Kaimana Beach

“Can we please get Biki actually in Kapiolani Park?”

— Lower Maikiki Biki User

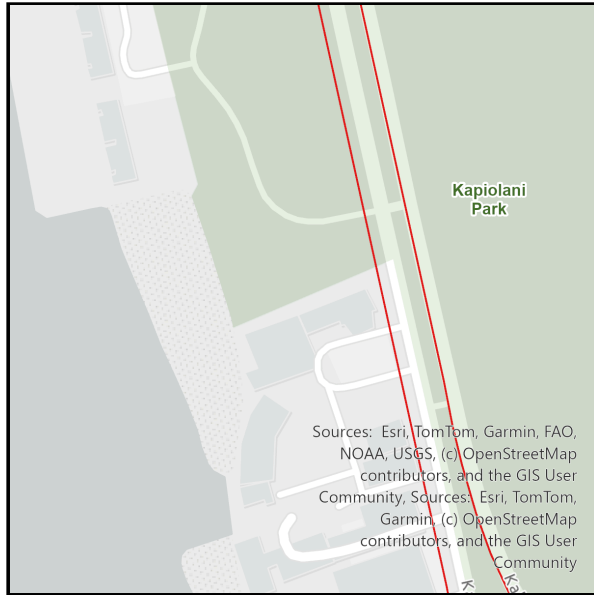


Figure 3: Kaimana Beach

For our next suitable location, we are seeking to extend the strip along the Waikiki beaches by adding a station that increases the connection between Kapiolani park, Kaimana beach hotel, and Diamond Head. In addition to these stations being among the most popular in the system, there is opportunity for good connection to bus stations in this area. We believe this station could be a good next step in shortening people’s first/last miles when traveling to the beaches, while also providing increased access to one of the best areas for recreational biking: Kapiolani Park

4 – Low-Contribution Stations

4.1 – Stations for Relocation

Using our ranking algorithm, we hand-picked the following stations for potential relocation. Low ranks for these stations indicate poor connection to nearby stations and limited incoming/outgoing traffic flow

Station ID	Station Name	Rank
83	403	0.05
34	400	0.052
45	455	0.071
48	234	0.072

Table 1: Lowest Ranking Selected Bike stations

4.2 – Ranking Limitations

External factors like bike may have unforeseen consequences on station ranking. For example, limited rebalancing might inhibit station usage. While these stations were ranked as some of the lowest contributing stations in your network, further investigation into each site is still necessary for validation, and a community survey on the relocation of each station is recommended (as noted in Section 2.4).

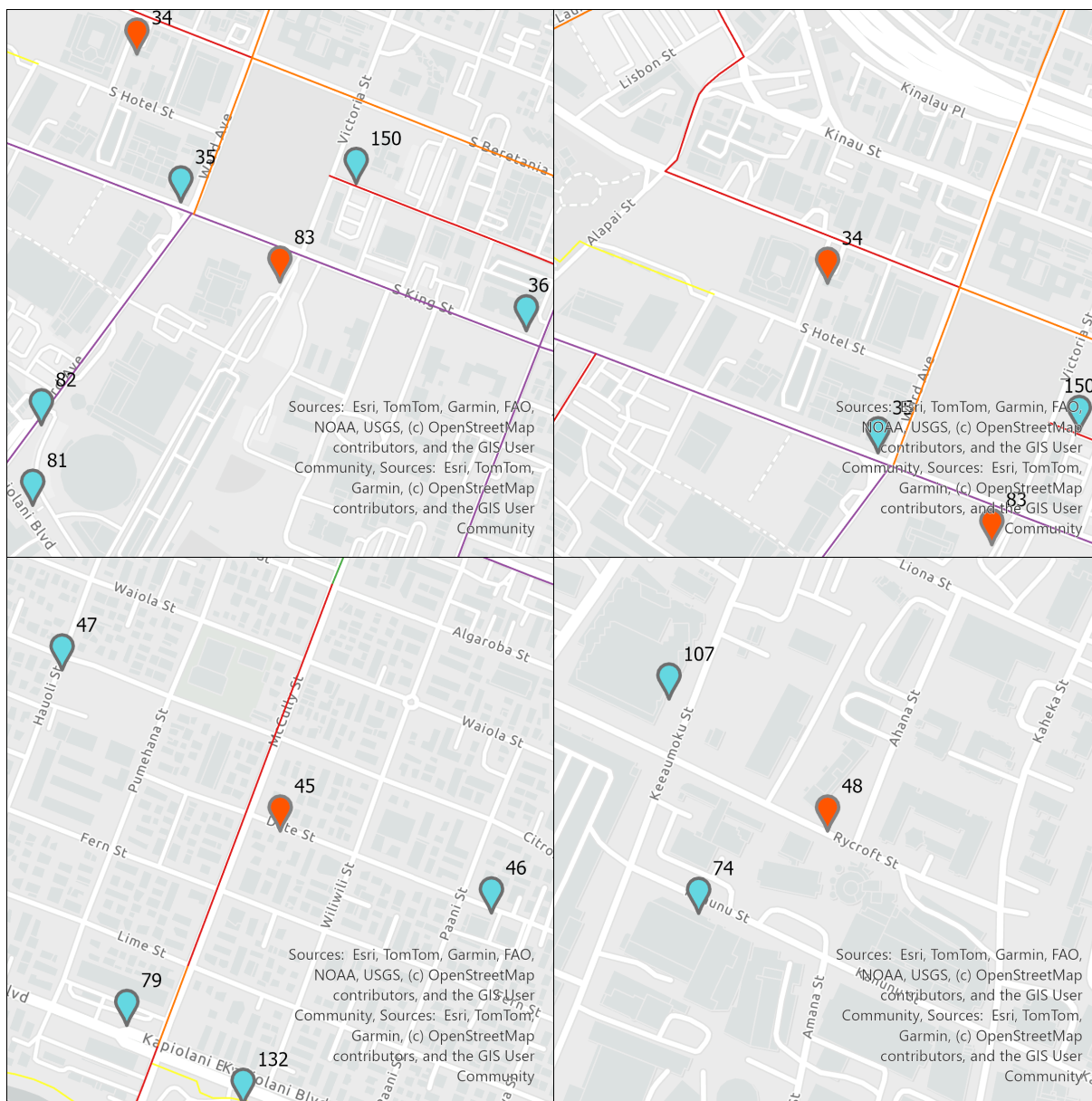


Figure 4: Map of candidate stations for relocation

5 – Long Term Recommendations

5.1 – Adding More Bikes

One of the largest barrier for bike sharing as reported by our survey was a lack of bikes and empty stations. While we recommend improved methods for rebalancing, a long-

term solution would combine increased rebalancing with the addition of more bikes to the network.

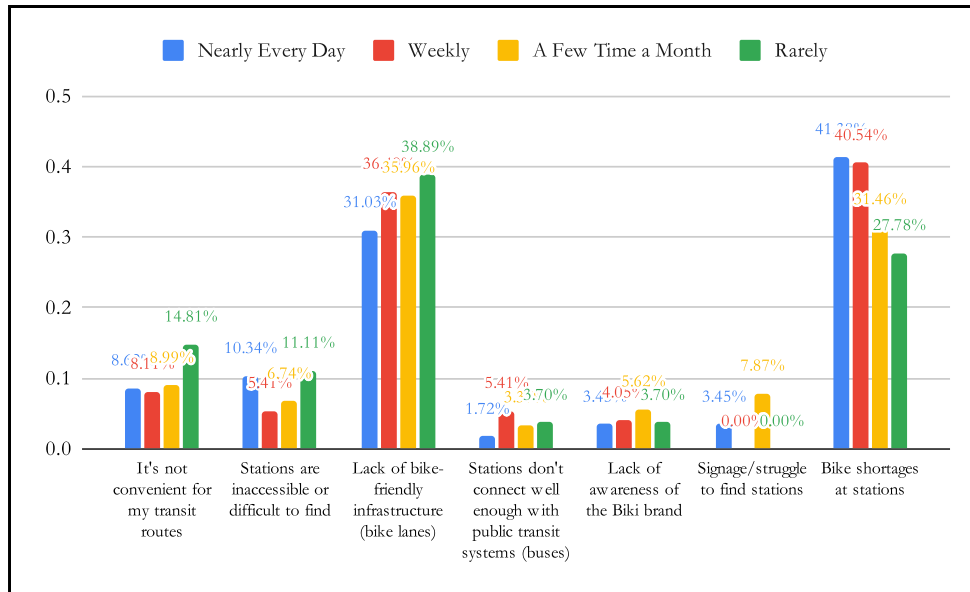


Figure 5: Largest Barriers to Biki Bike Usage Frequency (from Biki User Survey)

Biki Barriers conditioned on bike usage. The bars show that across bike riders of different usage frequencies, 27%-41% of users see bike shortages as a barrier to usage. The more frequently respondents use Biki, the more they perceive bike shortages as an issue

There are many different factors that determine how many bikes an ideal system should contain, here we will go over a few of them. We make the following assumptions for estimating the target number of bikes:

1. The area covered by Biki stations currently encompasses around 150,000 residents
2. There are about 775 bikes in Biki's system
3. Biki has about 130 stations with 2,275 individual docks

	Ideal ^b	Biki	New York ^c	Montreal ^c
Bikes per 1,000 Residents	10	5.3	6.3	6.1
Stations per sq. mi	25-40	17.6	30.7	15.5
Daily Trips per Bike	4-8	1.9	6.4 ^d	3.2 ^e
Docks per Bike^f	2-2.5	3.0	1.9	1.9

Table 2: Bike Fleet Statistics

Table 2 outlines various factors that contribute to the ideal flight size for a bikeshare system. Biki exhibits a low amount of bikes per 1,000 residents. However, this number is likely lower due to the large amount of tourist demand present in Honolulu.

^bbased on the IDTP Bike Share Planning Guide

^crough estimates based on published figures

^dNYC daily trips avg. is reduced by large electric bike demand reducing classic bike usage

^eMontreal doesn't operate all it's bikes during the winter, making peak summer use closer to 6.0

^fNYC & Montreal based on rough dock/station metric

Ratio	2.9	2.4	2.2	2.0
Amt. Of Bikes	775	950	1035	1140

Table 3: Amount of Bikes for Proposed Ratio

Table 3 demonstrates how many bikes Biki would need to contain to achieve that ratio of Docks/Bikes (with its current amount of docs). We suggest aiming for a ratio of 2.4 as it is within the IDTP recommendation.

5.2 – Docking Issues

Our survey received many reports about issues with bikes docking resulting in incorrect late fees. We understand that this is a known problem, but we believe that it’s a problem that deserves continual prioritization. Additionally, participants reported semi-frequent station outages with low resolution times. This can become a larger problem when coupled with low rates of bikes at stations.

Issues with reliability decrease convenience and will deter both old and new users. While this issue cannot be easily addressed, open communication with the public clarifying that you’re aware of the issues and working to resolve them can go a long way to earn back users’ trust.

5.3 – App Improvements

“...good to see more stations, but the app is not good. [Making it more] user friendly and [the] ability to see minutes remaining would be nice.”

— Kaka’ako / Ala Moana Biki User

Our survey showed some disapproval of the PBSC app, with many people preferring the old app. Some users spoke of how they had issues logging in or viewing their available credits. One respondent suggested adding an NFC functionality that they could use on their watch.

Our team has become aware of a potential solution with the new *Transit* app. We found that *Transit* had many of the same features as PBSC, while being more accessible and having built in integration with TheBus. Since it shows both bus routes and Biki stations, we believe that encouraging Biki users to start using *Transit* will help Biki with the rest of Honolulu’s infrastructure.

In addition to encouraging the use of *Transit*, we believe there are opportunities to further integrate Biki with it. If Biki moves forward with the transit app, there will be plenty of opportunities to add features using user feedback. Another example response from our survey is: “...once the bike is returned, a notification is sent to the registered email address immediately stating that the return has been accepted.”

5.4 – Bike Lanes and Integration

According to our survey, the another large perceived barrier towards people’s usage of Biki was the lack of bike-friendly infrastructure. From our experiences riding Biki, roads can often feel unsafe and make bike users feel unwelcomed. This issue can deter inexperienced riders from trying out Biki, cutting out a whole demographic of potential users. By continuing to work with the city to increase the safety and connectivity of bike lanes, we predict an increase of new bikers. Surveying users on locations for new bike lanes (see Section 2.4) is highly recommended.

In addition to working with the city to increase bike lane coverage, we think that it would an excellent idea to work on adding Holo card integration to Biki. We understand that this may already be in the works, so we want to encourage this as a way to make Biki a more accessible service. We also believe that incorporating Biki with the Holo card would give the government more incentive to provide funding for new bikes and repairs.

5.5 – Bike Accessories

Users of Biki have mentioned briefly in our survey that they would benefit greatly from having certain accessories on the bikes. These being phone holders and perhaps surfboard holders for a select few bikes. A specific quote from a user mentioned that there are “no surf racks on Biki (some would be ok).” A different user mentioned that they use Biki to “go surfing” in their survey response; adding surf racks to a small number of bikes for testing within Biki’s network would be great for the large surfing community in Honolulu.



Figure 6: An attachable surfboard rack accessory

5.6 – Maintenance Practices

A large issue that can be seen when taking out a Biki bike is the state in which they are left. Many users have commented on the quality of Biki’s bike maintenance, stating that they “want the sticky, worn-out handlebars to be fixed because they’ve deteriorated over time.” When grabbing a Biki bike, the first thing the user notices is the condition of the handles. Due to the climate of Hawai‘i, the rate of deterioration of these soft rubber handle grips happens much more quickly. If the regular replacement of handle grips is not viable, a material that is more resistant to UV, heat, and general wear would be highly recommended.

5.7 – E-bikes

Pulling from the open-ended responses of Biki’s user survey, we can see that there is a demand for incorporating e-bikes into Biki’s system. For example, some users responded saying that they “would love to see an expansion to the Biki area and the inclusion

of ebikes...”, as well as users mentioning that they “believe ebikes would be a great addition, especially for the hilly areas.”

Currently, there are plans for deploying e-bikes into Biki’s network; however, this addition will take time due to the need to install stations with charging capabilities, as well as having to purchase a suitable number of e-bikes for an initial rollout of the new system. Before expansion into steeper areas of Honolulu is considered, we recommend to first focus on adding e-bikes to Biki’s network. The pedal assist of Biki’s e-bikes would make these areas much more attractive to users and would give them a reason to bike uphill, rather than to leave bikes at stations of low elevation, hopefully helping with rebalancing as a result.



Figure 7: Biki’s current e-bike prototype

6 – Bus signage Info and Tracking

As a final note, we list our procedure to help gauge the effectiveness of the signage implemented into TheBus stations. The pairs are outlined in Table 4, along with their respective bracket and implementation.

In order to quantize the performance of bike stations near bus stop that have implemented signage, the following steps should be taken:

1. Calculate avg. monthly total rides for each bike stations.
2. Calculate monthly standard deviation of total rides across the previous year for each bike station.
3. For the Bus Stops with signs that had been implemented, if the avg monthly rides increases by a large percentage of the standard deviation from the avg of the previous year, consider that stop a successful implementation of the sign.
4. Do the same for the control stations, if there is similar trend disregard the previous results.

To build a better correlation between signage and bike ridership increase, obtain the bus arrival times for each station and the walking times between each bus station and it’s corresponding Biki station. Any outgoing rides from these stations within 1-10

(+ walking time) minutes after the corresponding bus arrives at its stop can noted. Compare these results to control Biki stations of similar size to observe whether or not the signs had a positive impact on ridership.

Bus Stop ID	Bike Station ID	Implemented/Control	Bracket
147	17	Implemented	0 - 1/16 Mile
145	77	Implemented	0 - 1/16 Mile
3296	11	Implemented	0 - 1/16 Mile
37	94	Implemented	0 - 1/16 Mile
883	52	Implemented	0 - 1/16 Mile
1102	151	Implemented	1/4 - 1/2 Mile
87	135	Implemented	1/4 - 1/2 Mile
95	14	Implemented	1/4 - 1/2 Mile
4688	37	Implemented	1/4 - 1/2 Mile
143	94	Implemented	1/4 - 1/2 Mile
363	132	Control Group	0 - 1/16 Mile
296	41	Control Group	0 - 1/16 Mile
4649	32	Control Group	0 - 1/16 Mile
433	81	Control Group	0 - 1/16 Mile
999	76	Control Group	0 - 1/16 Mile
126	165	Control group	1/4 - 1/2 Mile
309	127	Control group	1/4 - 1/2 Mile
24	24	Control group	1/4 - 1/2 Mile
1111	173	Control group	1/4 - 1/2 Mile
2086	38	Control group	1/4 - 1/2 Mile

Table 4: Bus Stop IDs Corresponding Bike IDs

7 – Final Notes

In an ideal world, we believe that the completion of this improvement plan would greatly increase Biki's ridership, bikeshare equity, and help close its first/last mile gap. However, we recognize that resources are limited, and that some of our recommendations could end up being quite costly. As we do not have the full picture of any finances, we would like to encourage you to follow this plan to the best of your ability without taking any actions that would put the bike share at risk.

Thank you,
WPI Bikeshare Team